

Session 2. Interpret

Workshop Exercises

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SESSION 2

Analysing AF3 outputs using ChimeraX

Basic commands, visualizing confidence metrics, and exporting interface data.

Session objectives

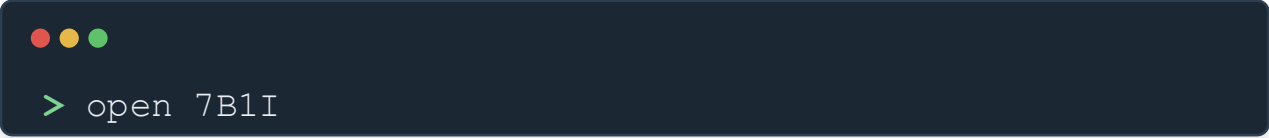
Objectives

The main goal of this session is to become familiarized with AF3 output analysis, using commands in the molecular visualization program UCSF ChimeraX.

EXERCISE 2.1

Introduction to ChimeraX basic commands

Open a protein model from the Protein Data Bank (PDB), make chain and residue selections, and color different regions. To begin, run this command in the ChimeraX console:

A dark-themed terminal window with three colored window control buttons (red, yellow, green) in the top-left corner. The text "> open 7B1I" is displayed in a light green monospace font.

```
> open 7B1I
```

In the log, click on [\[more info...\]](#) to look at the metadata associated with 7B1I.

?

METADATA

- a) What's the title of the paper associated with 7B1I?
- b) What was the experimental method used for its resolution?

EXERCISE 2.1

Chains, sequences and secondary structure



Navigate to the **Molecule Display** panel and click the **chain** icon.



Now click the **Sequence** icon.



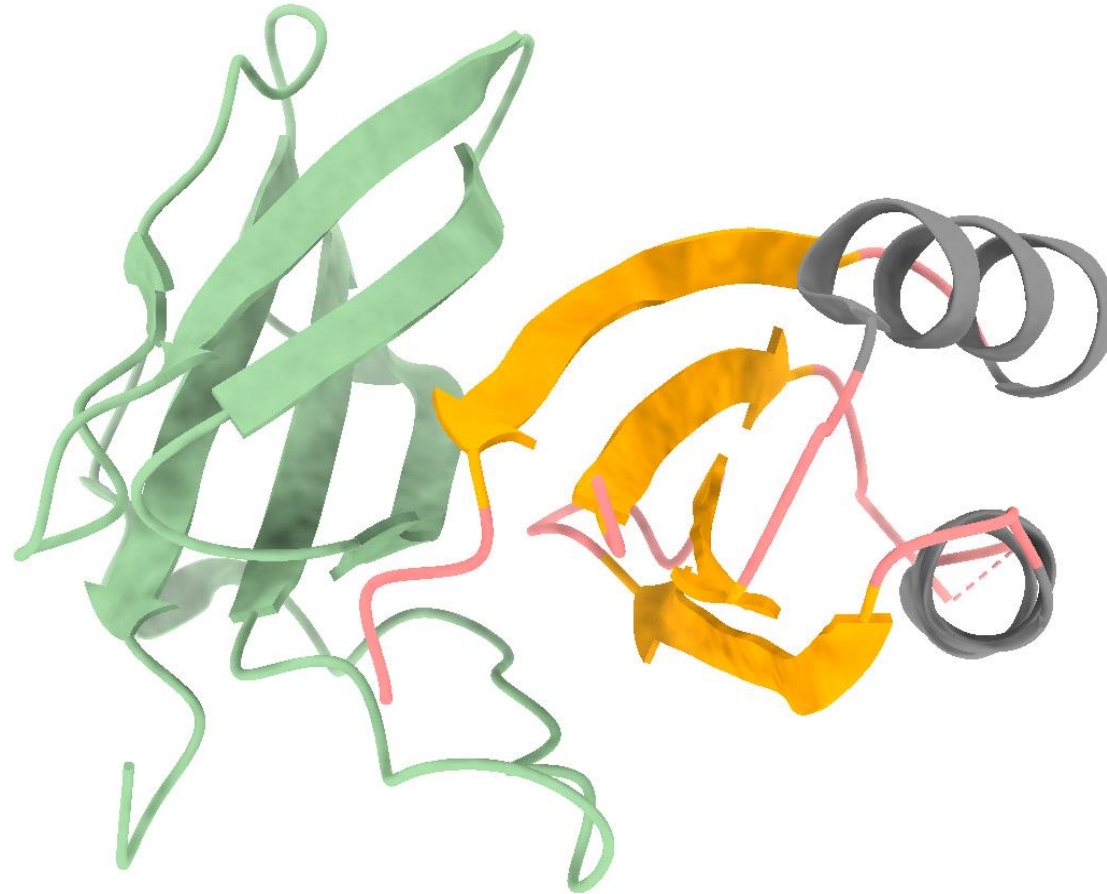
DISCUSSION

c) Which chain in the model is longer, chain A or B?

Color all α -helices gray: in the sequence panel for chain B, click one of the yellow rectangles (α -helices) to select all α -helices on chain B, then use **Actions** → **Color** → **Gray**. Now do the same for the β -sheets and color them **orange**. Your model should now look like the figure on the next slide.

EXERCISE 2.1

7B1I coloured by secondary structure



Chain B with α -helices in gray and β -sheets in orange.

EXERCISE 2.1

The same, from the ChimeraX console

The **#** denotes the model ID (we only have model '1' open) and the slash **/** specifies the chain ID.



```
> sel #1/c    select chain C  
> color sel gray  color the selection gray
```

Selection and coloring can be condensed into one line:



```
> color #1/c #2B89AD  hex code for a teal blue
```

Tip *Using a hex code (e.g. #2B89AD) is convenient when you need a color that is not one of the default options.*

EXERCISE 2.1

Selecting residue ranges and lists

Specify a range or list of residues using a colon `:` after the chain ID.

Hierarchical specifiers used by ChimeraX:

Symbol	Reference Level	Definition	Examples
#	model	model number assigned to the data in ChimeraX (hierarchical, with positive integers separated by dots: N, N.N, N.N.N, etc.)	#1 #1.3
/	chain	chain identifier (case-insensitive unless both upper- and lowercase chain IDs are present)	/A
:	residue	residue number OR residue name (case-insensitive)	:51 :glu
@	atom	atom name (case-insensitive)	@ca



```
> sel #1/b:1-10    residues 1 to 10
> sel #1/b:1,5,20  residues 1, 5 and 20
> sel #1/b:1-10,66-73  two ranges
```

Now colour each β -sheet of the protein:



```
> color #1/b:3-9 blue     $\beta$ -sheet no. 1
> color #1/b:30-36 green  $\beta$ -sheet no. 2
> color #1/b:42-47 orange  $\beta$ -sheet no. 3
> color #1/b:66-73 red    $\beta$ -sheet no. 4
```

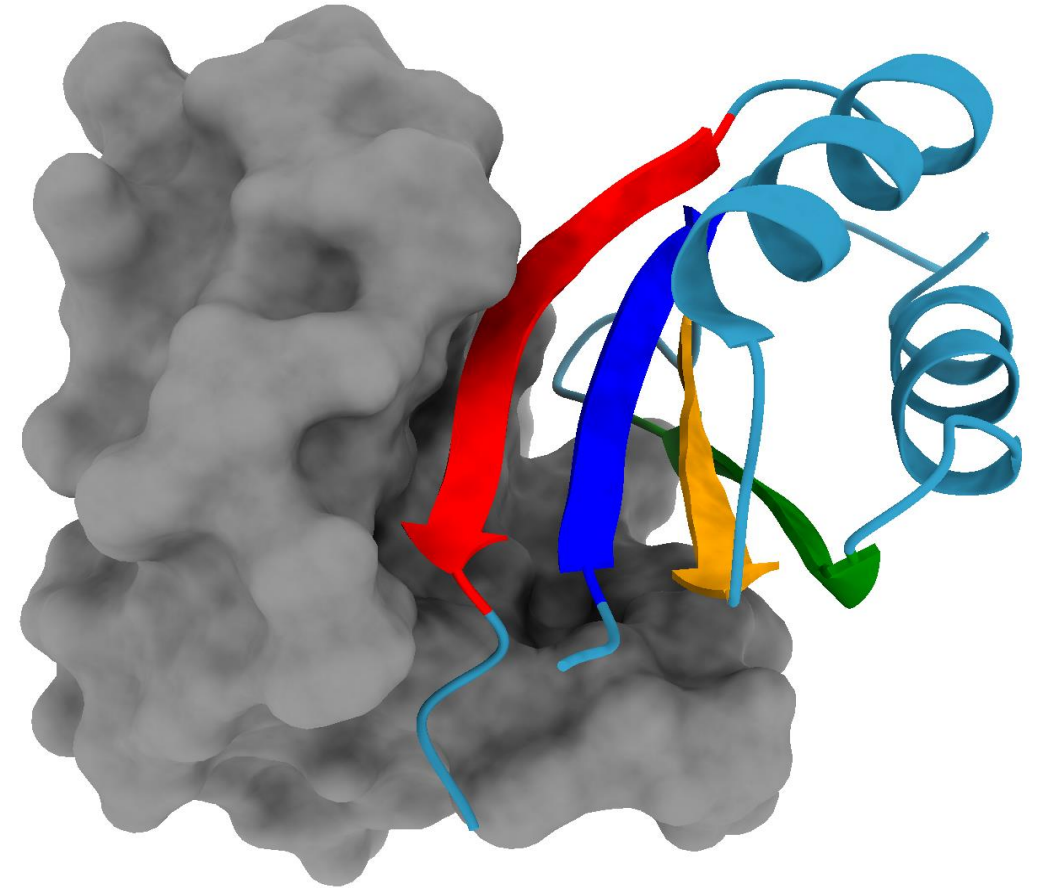
EXERCISE 2.1

Showing a molecular surface

Finally, show the AVR-Pik protein (chain C) in volumetric representation by displaying its surface:

```
> surface #1/c
```

Find an angle that shows the binding of AVR-Pik to OsHIPP19, then take a picture and a spin movie from the **Home** panel → **Images** subpanel.



AVR-Pik (surface) bound to OsHIPP19 (ribbon).

EXERCISE 2.1

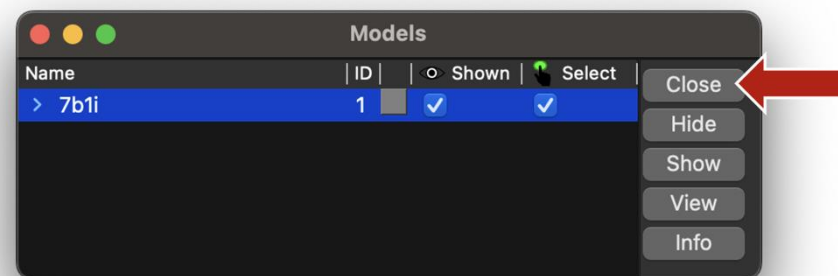
Saving your session and closing the model

By default the picture and video are saved to your Desktop. Check where they were saved by typing `pwd` (print working directory) in the console — the directory is printed in your log.

ChimeraX does not save progress automatically. Save a session by typing the command (*the direction of the '/' bars depends on your operating system*):

```
> save ~/Users/{username}/Desktop/  
ChimeraX_Exercise1.cxs
```

Once saved, select the 7B1l model in the **Models** panel and click **Close**.



Select the model, then click Close.

EXERCISE 2.2

Visualizing AF3 models with their confidence metrics

We will visualize AlphaFold 3 prediction results in ChimeraX along with their confidence metrics — colouring residues by **pLDDT** scores and displaying **PAE** values for contacting residues as pseudobonds.

We will use the prediction from Exercise 1: Pikip-1_186–486 in complex with Pathogen_Protein_1. Download the results from the AF3 web server (Download button). You should obtain a ZIP file, e.g.:

```
fold_pikip_1_186_486_with_pathogen_protein_1.zip
```

Unzip the file to extract the contents — you should see 17 files, including:

- [...]_model_0.cif — top-ranked model, with the predicted atomic coordinates
- Several .cif files for the other ranked models
- .json files containing the PAE data (error estimates) for each model

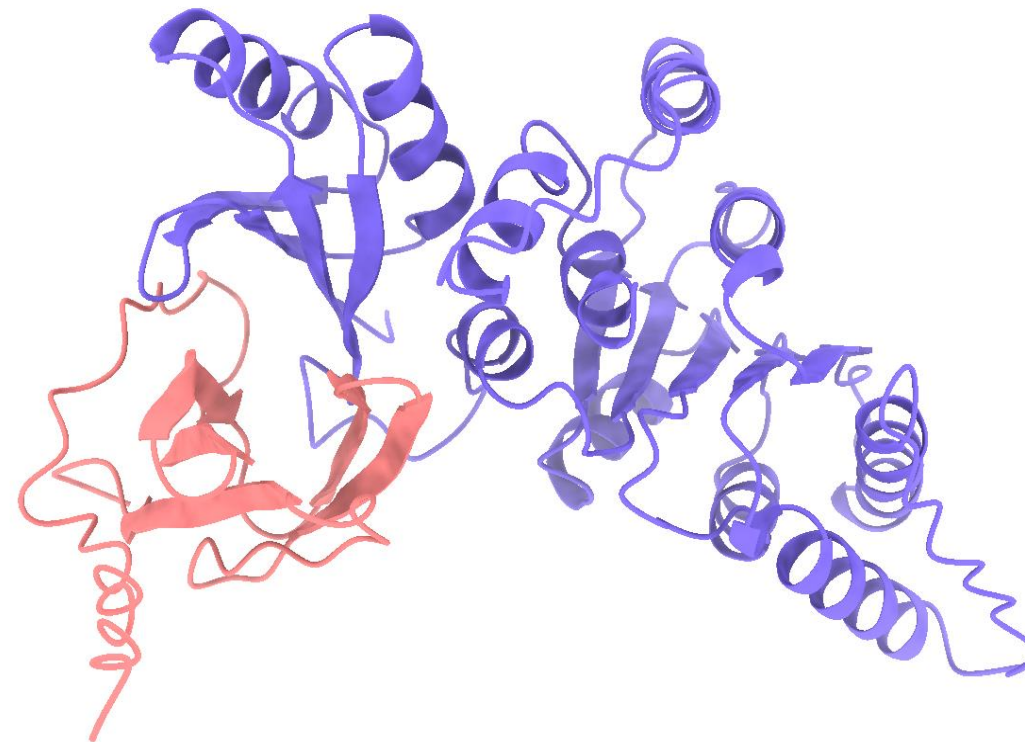
EXERCISE 2.2

Load the model and colour by chain

We will analyze the top-ranked model — `model_0.cif`. Drag this file into ChimeraX to load the structure, then colour by chains:

```
> color #1 bychain
```

Show the sequences (Sequence icon). From the sequences it is evident that the shorter **chain B** is the effector Pathogen_Protein_1, and the longer **chain A** is Pikp-1_186–486.



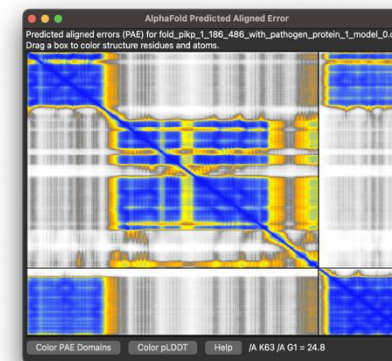
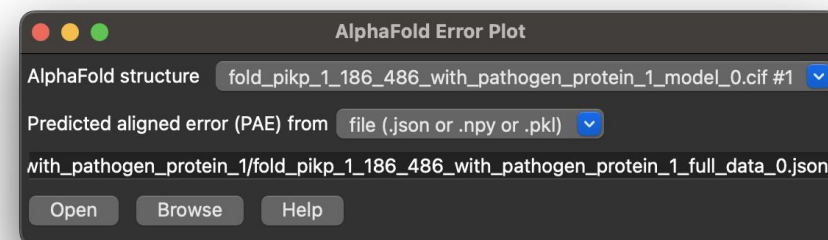
model_0.cif coloured by chain (chain A vs chain B).

EXERCISE 2.2

Load the confidence metrics (PAE)

1. In the top menu, go to: Tools → Structure Prediction → AlphaFold Error Plot
2. A new window opens with a dropdown to select the model.
3. In most cases ChimeraX finds the correct .json automatically. If not, click Browse and select the JSON for your top-ranked model (..._full_data_0.json), then Open.

You should now see the PAE data for your model. This window shows the error data and lets you colour the structure by **pLDDT** scores or by **PAE domains**.

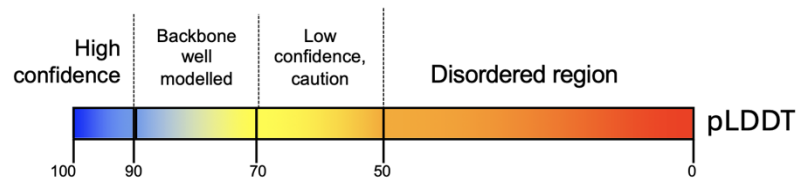
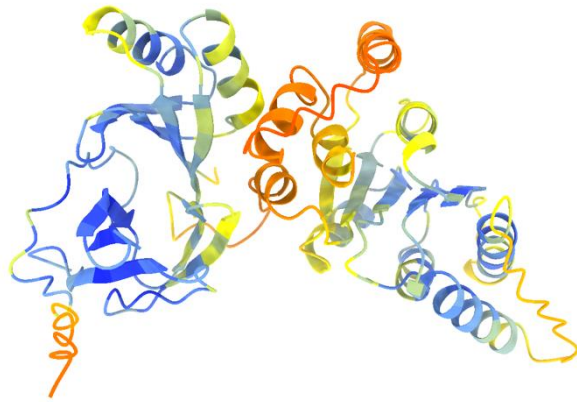


AlphaFold Error Plot window and PAE data.

EXERCISE 2.2

Colour the prediction by pLDDT

Colour the prediction by pLDDT — you should get something like this:



?

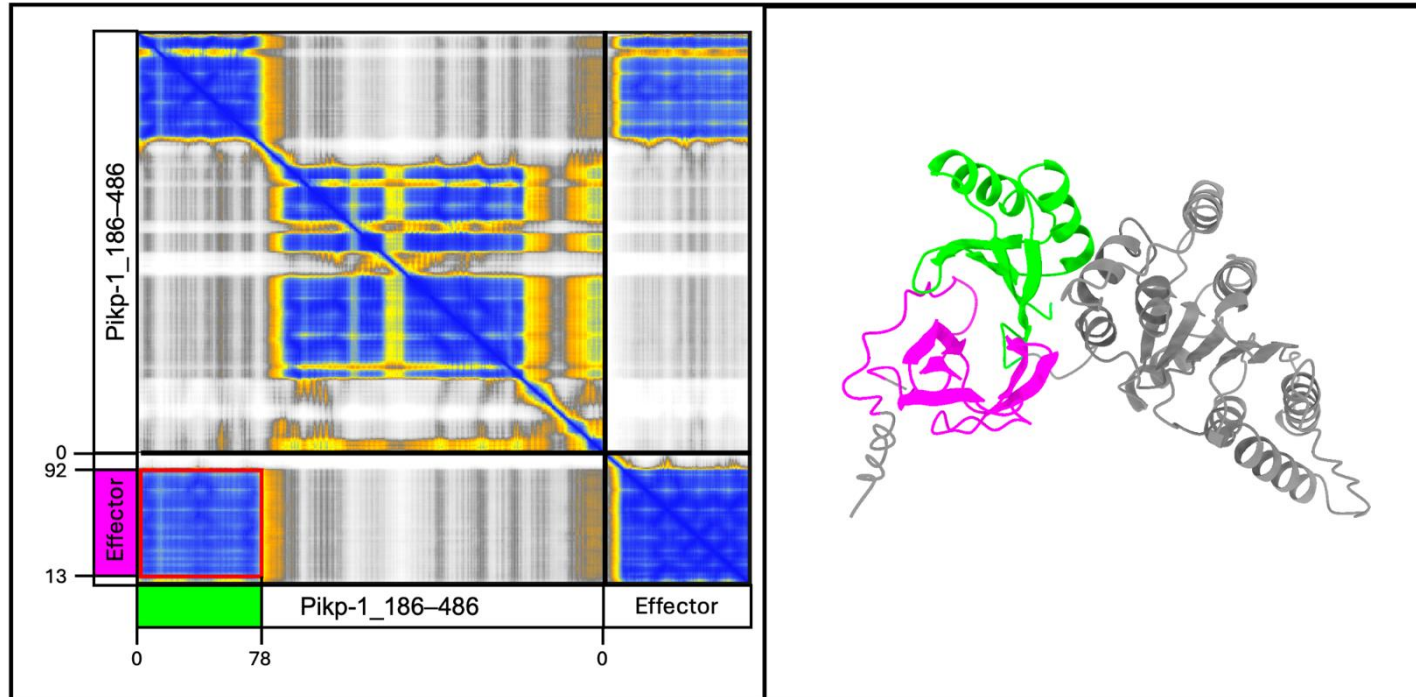
DISCUSSION

A) What can you say about the pLDDT scores of the residues predicted at the interaction interface between the two proteins?

EXERCISE 2.2

The PAE plot is interactive

The PAE plot is interactive: a selection in the plot is reflected in the 3D model. Selecting the high-confidence region in the marginal highlights (in pink and green) the **N-terminal end of Pikp-1_186–486** and almost all of the effector. AF3 is confident about the relative positioning of the Pikp N-terminal residues with respect to all effector residues.



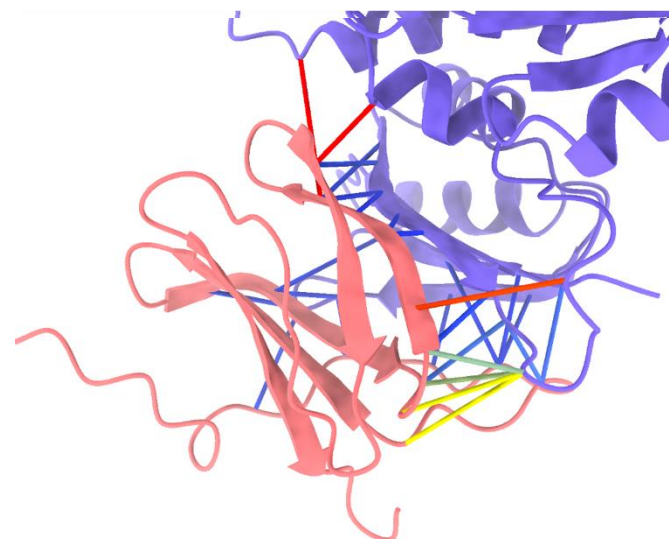
EXERCISE 2.2

Visualizing predicted contacts as pseudobonds

Another way to visualize PAE for predicted contacts (requires PAE data loaded):

```
> alphafold contacts #1/a to #1/b distance 3.5
```

Any residue pairs within 3.5 Å are joined by a 'pseudobond', coloured by the AlphaFold PAE metric using the ChimeraX **paecontacts** palette.



Interface pseudobonds coloured by PAE.



Predicted Aligned Error (PAE)

Reading and refining the contacts



DISCUSSION

B) Can you tell which β -sheet on the side of the effector protein is predicted to have more high-confidence (i.e. low PAE) contacts with Pikip, at a distance ≤ 3.5 Å?

The command can also target specific ranges or lists of residues, using the same selection algebra as Exercise 2.1.



```
> alphafold contacts #1/b:20 to #1/a distance 4.5
```



DISCUSSION

C) If we run the command above, we will be plotting contacts between which residues, and at what distances?

EXERCISE 2.3

Exporting information from a protein–protein interface

Export the predicted contacts from Exercise 2.2 into another program (e.g. Excel) for downstream analysis. Export the list of residues to a text file by adding **outputFile** to the command:

```
> alphafold contacts #1/a to #1/b distance 3.5 outputFile AF3-contacts.txt
```

The result is a text file where each row is a pair of residues in contact below the 3.5 Å cutoff, with a given PAE score. This data can be further processed in Excel to remove low-confidence contacts (e.g. /A:81 /B:54 17.50).

PAE confidence 0–10 = confident · 10–15 = gray zone · >15 = poor confidence

EXERCISE 2.3

From ChimeraX to Excel

```
AF3-contacts.txt — Edited
/A:71 /B:56 1.90
/A:33 /B:25 1.70
/A:35 /B:23 1.80
/A:74 /B:53 2.60
/A:75 /B:50 4.10
/A:75 /B:28 3.90
/A:3 /B:28 3.40
/A:69 /B:63 2.50
/A:72 /B:56 1.60
/A:68 /B:58 2.20
/A:93 /B:63 19.70
/A:77 /B:53 6.70
/A:77 /B:50 7.20
/A:77 /B:32 9.70
/A:77 /B:51 9.30
/A:47 /B:25 1.60
/A:81 /B:54 17.50
/A:38 /B:21 1.50
/A:38 /B:45 1.90
/A:43 /B:45 1.80
/A:39 /B:45 1.50
/A:39 /B:21 1.60
/A:39 /B:18 2.20
/A:39 /B:43 2.00
/A:70 /B:58 1.70
/A:70 /B:57 1.50
/A:45 /B:25 1.60
/A:278 /B:58 24.60
```



Position A (Pikp-1_186-486)	Position B (Pathogen_Protein_1)	PAE
/A:38	/B:21	1.5
/A:39	/B:45	1.5
/A:70	/B:57	1.5
/A:39	/B:21	1.6
/A:45	/B:25	1.6
/A:47	/B:25	1.6
/A:72	/B:56	1.6
/A:33	/B:25	1.7
/A:70	/B:58	1.7
/A:35	/B:23	1.8
/A:43	/B:45	1.8
/A:38	/B:45	1.9
/A:71	/B:56	1.9
/A:39	/B:43	2
/A:39	/B:18	2.2
/A:68	/B:58	2.2
/A:69	/B:63	2.5
/A:74	/B:53	2.6
/A:3	/B:28	3.4
/A:75	/B:28	3.9
/A:75	/B:50	4.1
/A:77	/B:53	6.7
/A:77	/B:50	7.2
/A:77	/B:51	9.3
/A:77	/B:32	9.7
/A:81	/B:54	17.5
/A:93	/B:63	19.7
/A:27	/B:58	24.6

Exported AF3-contacts.txt processed in Excel — contacts sorted by PAE, coloured green (confident) to orange (poor).

Design and test interface mutations



DISCUSSION

- A)** Study the predicted interface of Pathogen_Effector_1 and rice Pikp-1_186–486, and generate a list of candidate residues (in either protein) that you would mutate to abolish their interaction.
- B)** What key experiments would you conduct to test these designed mutants in the lab? Briefly describe how you would design the experiments, and which controls you would incorporate.